

IMMUNOLOGY

Deep learning reveals predictive sequence concepts within immune repertoires to immunotherapy

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T cell receptor (TCR) sequencing has been used to characterize the immune response to cancer. However, most analyses have been restricted to quantitative measures such as clonality that do not leverage the complementarity-determining region 3 (CDR3) sequence. We use DeepTCR, a framework of deep learning algorithms, to reveal sequence concepts that are predictive of response to immunotherapy. We demonstrate that DeepTCR can predict response and use the model to infer the antigenic specificities of the predictive signature and their unique dynamics during therapy. The predictive signature of nonresponse is associated with high frequencies of TCRs predicted to recognize tumor-specific antigens, and these tumor-specific TCRs undergo a higher degree of dynamic changes on therapy in nonresponders versus responders. These results are consistent with a biological model where the hallmark of nonresponders is an accumulation of tumor-specific T cells that undergo turnover on therapy, possibly because of the dysfunctional state of these T cells in nonresponders.

INTRODUCTION

The advent of checkpoint inhibition [α -programmed death 1 (α -PD1) and α -cytotoxic T lymphocyte associated protein 4 (α -CTLA4)] in cancer has changed the landscape of how oncologists understand and treat advanced and aggressive disease (1–3). The initial trials of checkpoint inhibition or blockade demonstrated the possibility for obtaining durable clinical benefit in advanced metastatic cancer (4–6); however, while there has been tremendous promise of immunotherapy, most patients still do not respond to this mode of treatment (6). Thus, great attention has been made to the development of predictive biomarkers to select for which patients may benefit the most from immunotherapy and to further our understanding of successful use of checkpoint blockade in cancer (7–20).

One promising area of interest has been examining the T cell response through broad analysis of the T cell repertoire via T cell receptor sequencing (TCR-seq), where the TCR sequence is obtained through next-generation sequencing, allowing for a characterization of the antigenic determinants of response (12, 13, 21–23). However, the vast majority of the work in this area has been limited to characterizing the diversity of the TCR repertoire (either at the clonotype or structural level) (24–32). Previous work has demonstrated that certain sequence concepts can predict tumor types (33) but has yet to be applied in an effort to predict response to immunotherapy, and thus, we wanted to query whether there were sequence concepts (i.e., motifs) that were predictive of response to immunotherapy.

We used the repertoire classifier from DeepTCR, a previously described set of deep learning algorithms, to search for sequence concepts that were predictive of response to immunotherapy (22, 23).

We applied DeepTCR to a clinical trial dataset (CheckMate-038) where patients with inoperable melanoma received either α -PD1 or α -PD1 + α -CTLA4, were sequenced before and after initiation of therapy and where the ground truth label corresponds to radiographic response via RECIST v1.1 criterion. This analysis not only represents the value of DeepTCR as a pre-treatment biomarker predictive of response in a clinical immunotherapy setting, reanalyzing recently published TCR-seq data from Anagnostou *et al.* (21), but we further use this predictive modeling to reveal biological insights into the antigenic determinants of response and their dynamics under the influence of immunotherapy.

RESULTS

Extending DeepTCR's repertoire classifier with human leukocyte antigen

We first expand our previously described multiple-instance learning (MIL) repertoire classifier by allowing for incorporation of the human leukocyte antigen (HLA) into the featurization of the TCR to provide a representation of a joint TCR-HLA antigenic latent space (Fig. 1). This is particularly important when analyzing human-derived data as individuals often have different HLA backgrounds and their tumors have unique mutation-associated neoantigens and tumor-associated antigens; thus, TCRs that are structurally homologous would likely not be recognizing the same tumor antigens if being compared from patients with different HLA backgrounds. By incorporating HLA into the featurization of the TCR, we provide contextualization of the TCR by the HLA background it was observed within and thus allow direct comparisons between individuals/patients with differing HLA backgrounds.

DeepTCR's repertoire classifier is predictive of response

We first applied DeepTCR to the pre-treatment tumor specimens from the cohort of patients enrolled in the CheckMate-038 clinical trial (fig. S1). Pre-treatment tumor biopsies were collected and TCR-seq was performed from 43 patients enrolled in the CheckMate-038 (parts 2 to 4) clinical trial where they were either treated with α -PD1 monotherapy (9 patients) or α -PD1 + α -CTLA combination therapy

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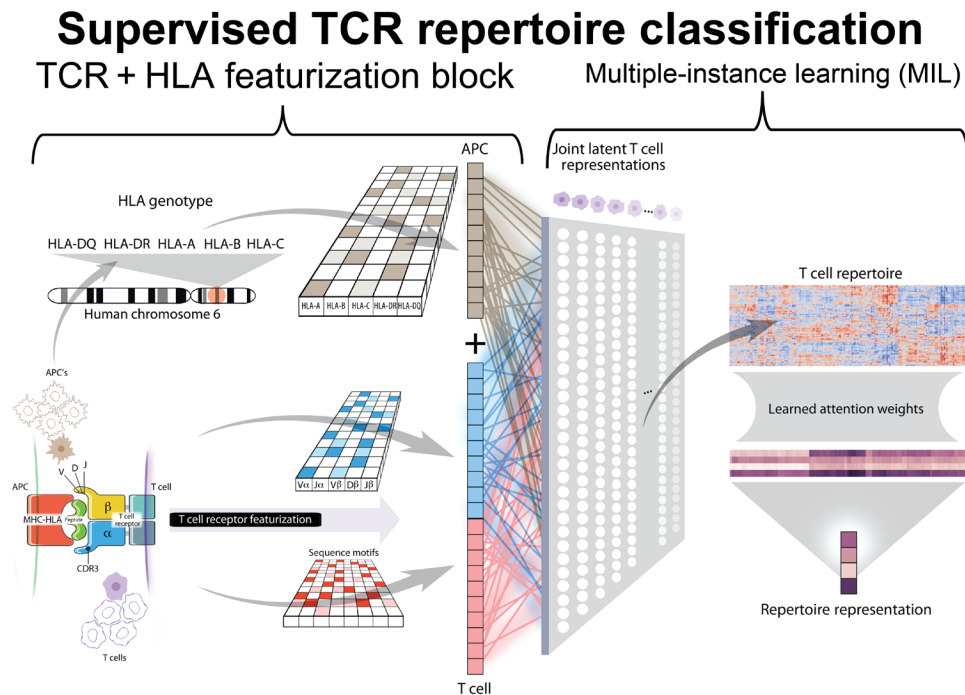


Fig. 1. HLA-extended, MIL-supervised TCR repertoire model. We expand on previous work by modifying the TCR featurization block to incorporate the HLA background within which a given collection of TCRs were observed. The HLA background of a sample/individual is provided to the neural network in a multi-hot representation that is re-represented in a learned continuous embedding layer and concatenated to the continuous learned representation of the TCR. As previously described, we implement a multihead attention mechanism to make sequence assignments to concepts within the sample. The number of concepts in the model is a hyperparameter, which can be varied by the user depending on the heterogeneity expected in the repertoires. Notably, this assignment of a sequence to a concept is done through an adaptive activation function that outputs a value between 0 and 1, allowing the network to put attention on the sequences that are relevant to the learning task. When taking the average of these assignments over all the cells in a repertoire, this results in a value within the neural network that directly corresponds to the proportion of the repertoire that is described by that learned concept. These proportions of concepts in the repertoire are then sent into a final traditional classification layer. Adapted from Sidhom *et al.* (22), T. Phelps, Department of Art as Applied to Medicine, The Johns Hopkins University School of Medicine. APC, antigen-presenting cell; MHC, major histocompatibility complex.

(34 patients) and followed for radiographic response to therapy via RECIST v1.1. Complete responders and partial responders (CRPR) were denoted as responders to therapy while stable disease and progressive disease (SDPD) were denoted as nonresponders to therapy. We observed that when applying DeepTCR to predict response to immunotherapy, the joint representation of TCR with HLA outperformed models of DeepTCR that used TCR sequence or HLA genotype information alone [area under curves (AUCs); TCR = 0.77, HLA = 0.75, TCR + HLA = 0.86, random permutation testing = 0.515; Fig. 2A, stratified by treatment cohort in fig. S2]. The resulting DeepTCR's TCR + HLA model predictions of likelihood to respond to treatment also significantly stratified progression-free survival (PFS) in this cohort of patients (Fig. 2B). The TCR + HLA model exhibited less variability during cross-validation assessments as compared with either the TCR or HLA model alone (fig. S3), and it appears that the addition of HLA genotype conditions the model to learn different information from the TCR sequence data (fig. S4). We also found that in this cohort, the DeepTCR TCR + HLA model performed comparably to conventional biomarkers that have been previously described to stratify response in immunotherapy-treated patients [AUCs; programmed death ligand 1 (PD-L1; immunohistochemistry tumor proportion score) = 0.693, total mutational burden (TMB; exome based) = 0.772, TCR clonality = 0.642, and total T cell count = 0.825; fig. S5]. In addition, while similar performance

(based on AUC) is observed with total T cell counts (a surrogate of the degree of T cell tumor infiltration) and DeepTCR, these predictors are independent of multivariate logistic regression analyses and as such plausibly represent complementary information. To further validate our findings, we collected two independent validation cohorts of patients (denoted as yost and sade) with skin cancer who were treated with checkpoint blockade and underwent either bulk TCR-seq or single-cell TCR-seq before initiation of treatment (34, 35). When using the 100 models fit in Monte Carlo (MC) cross-validation on the CheckMate-038 dataset in ensemble on these two cohorts of patients, we found the predictive signature of response in varying degrees in both these cohorts before initiation of treatment [AUCs: yost = 0.82 ($n = 11$), sade = 0.61 ($n = 19$), and yost + sade = 0.71 ($n = 30$)], providing further validation of this predictive signature of response to checkpoint blockade (Fig. 2C).

Unsupervised representation reveals nature of predictive antigenic response

To characterize the distribution of the TCR repertoires in patients who either responded or did not respond to treatment, we trained a variational autoencoder (VAE), another type of model part of the DeepTCR framework (22), on all data to obtain an unsupervised featurization to visualize [via uniform manifold approximation and projection (UMAP)] the distribution of nonresponders and responders

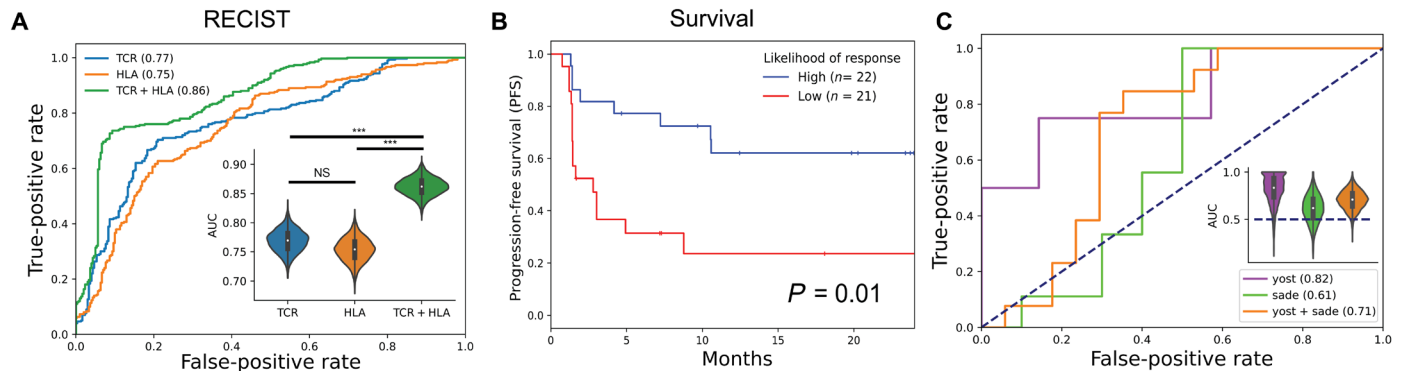


Fig. 2. Repertoire classification in human pre-treatment tumor-infiltrating lymphocytes. (A) Receiver operating characteristic (ROC) curves were created for predicting response (complete response and partial response) to immunotherapy given TCR, HLA, or TCR + HLA information to the supervised repertoire classifier [100 Monte Carlo (MC) simulations with a train size of 37 and a test size of 6]. Bootstrap analyses (5000 iterations) were performed to construct confidence intervals (CI) around AUC values and assess differences in model performance, in which each AUC per sampling was compared in a paired manner across the three models designed above. The null hypothesis of two models exhibiting equivalent performance was rejected if the bootstrap difference did not cross 0 (***, 99.9% CI). NS, not significant. (B) The likelihood of response generated by the TCR + HLA model was dichotomized into “high” and “low” using the median predicted value in this cohort (taken over the MC test sets and averaged per sample) and the Kaplan-Meier curves were shown for PFS, log-rank $P = 0.005$. (C) Repertoire models fit on CheckMate-038 data were used to conduct repertoire level ensemble inference on two previously published cohorts of patients with skin cancer undergoing treatment with checkpoint blockade. Performance characteristics are shown via ROC curves for both cohorts independently and combined [yost ($n = 11$), sade ($n = 19$), and yost + sade ($n = 30$)]. Bootstrap analyses, as previously described, were performed to construct CIs around AUC values.

(Fig. 3A; per sample distributions shown in fig. S6). While no difference in the overall distributions can be appreciated when looking at all the data, we found that when filtering the TCR sequences to the top and bottom 10% predictive sequences (Fig. 3B), we were able to visualize differences in the TCR repertoires between responders and nonresponders (Fig. 3C), illustrating the functionality of the MIL algorithm to “denoise” TCR repertoire. We noted that not only are the distributions within responders and nonresponders multimodal but also these multiple modes are shared between patients (Fig. 3D).

Predictive signature persists while on therapy

We next wanted to query whether this predictive signature of response (pre-tx) persisted in the TCR repertoires taken post-treatment (post-tx). To do this, we applied the models (TCR, HLA, and TCR + HLA) trained on the pre-treatment data to 35 paired post-treatment samples and examined the receiver operating characteristic (ROC) curves and AUCs between pre-treatment and post-treatment (Fig. 4A). When doing this, the results demonstrated comparable performance between the pre-treatment and post-treatment samples, suggesting that the predictive signature was still present even weeks after initiating immunotherapy. Furthermore, we noted a high correlation between the pre-treatment and post-treatment prediction values (Fig. 4B), suggesting maintenance of pre-treatment response-relevant repertoires during treatment. When comparing the 35 paired pre- and post-treatment repertoires, examining the top 10% predictive sequences for each class (i.e., CRPR versus SDPD) either at the sample level (Fig. 4, C and E) or entire cohort level (Fig. 4, D and F), we demonstrated a conserved predictive antigenic response while on therapy. Together, these findings suggest that the antigenic response is not only different between responders and nonresponders to checkpoint blockade, but each cohort of patients contains a set of T cell responses that likely recognize a broad class of structurally related antigens and these predictive concepts persist while on treatment.

Predictive signature of nonresponse is associated with tumor-specific TCRs

To further characterize the antigenic specificity of the antigenic response, we first created residue sensitivity logos as described in the original DeepTCR publication (22) of the top 50 most predictive TCRs of responders (CRPR) and nonresponders (SDPD) (Fig. 5A). We noted that the most predictive residues were localized to the central portion of the sequence, suggesting that the predictive signature was indeed related to the antigen specificity of the TCR. We then used a previously published dataset in melanoma by Oliveira *et al.* (36) where the authors had paired TCR sequence data with known specificities (i.e., viral versus neoantigen versus tumor-associated antigens), along with the known HLA background of the individuals from which the sequences were obtained. We took these TCRs and ran them through the previously trained DeepTCR repertoire classifier to assign per-sequence likelihoods of response versus nonresponse. We noted that when looking at the distribution of likelihoods across the various types of antigens, the virus-specific TCRs [Epstein-Barr virus (EBV), Influenza (Flu), and yellow fever (YF)] had a higher likelihood of response than the tumor-specific TCRs [NeoAg and melanoma antigen recognized by T cells 1 (MART-1)] (Fig. 5, B and C). To further validate these findings, we pulled TCRs from the McPas-TCR database (37) and crossed exact sequence matches with the TCRs found within the CheckMate-038 cohort and then looked at the corresponding likelihood of response of these cross-matched TCRs. Once again, we noted a similar finding to the dataset of Oliveira *et al.* (36) in that the virus-specific TCRs (EBV, cytomegalovirus, flu, and YF) had a higher likelihood of response than the tumor-specific TCRs (MART-1) (Fig. 5D). When visualizing these virus-specific versus MART-1-specific TCRs in the unsupervised TCR sequence space, we also saw that the virus-specific TCRs were more enriched in the responder-specific areas and the MART-1-specific TCRs were more enriched in the nonresponder-specific areas of the UMAP (Fig. 5E).

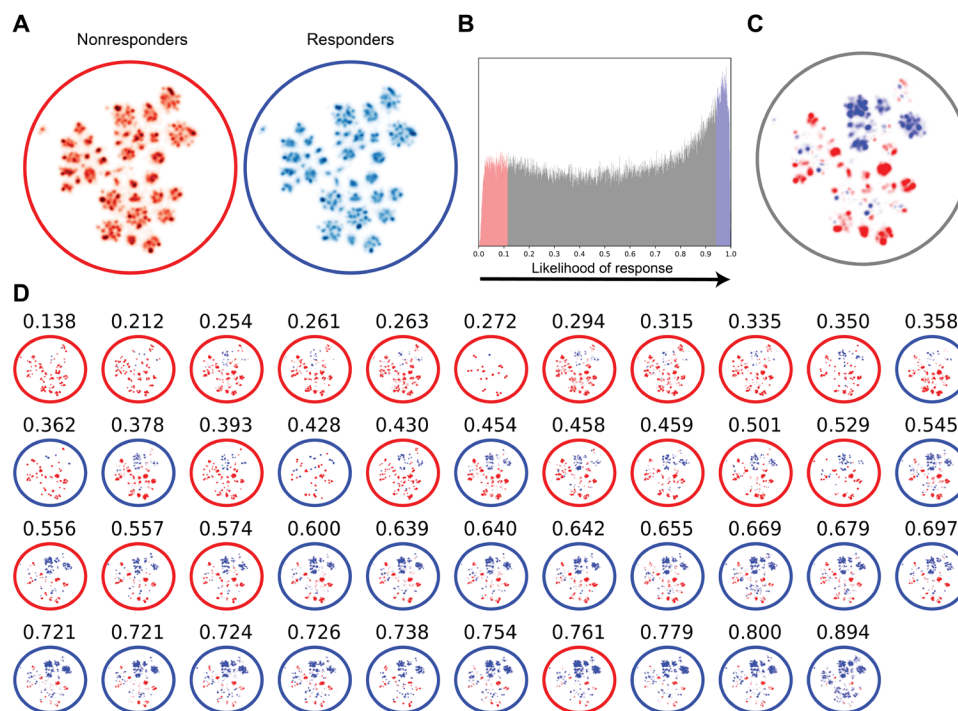


Fig. 3. Unsupervised representation of predictive signature. (A) To provide a descriptive understanding of the T cell response in responders and nonresponders in the CheckMate-038 clinical trial, we sought to characterize the distribution of the TCR repertoire in this cohort of patients. Data from CheckMate-038 were used to train a VAE on all sequence data (incorporating TCR + HLA information) in a sample and class agnostic fashion. The distribution of responders and nonresponders repertoires was visualized via UMAP of the unsupervised VAE featurization (per sample distribution shown in fig. S6). (B) To visualize the distribution of the highly predictive TCR sequences, a per-sequence prediction value was assessed following each MC simulation on the TCR's within the independent test set, assigning the probability that a given TCR had a responder signature. Over the 100 MC simulations, each sequence in the cohort is assigned multiple prediction values that are averaged over all simulations to serve as a consensus predicted value for each sequence in this cohort of patients. (C) Top 10% of sequences in responders and nonresponders were selected and visualized over the entire cohort and on a per-sample basis (D) where edge color denotes the ground truth label of the sample (nonresponder, red; responder, blue) and average predicted likelihood taken over MC simulations to respond to treatment shown above each patient's distribution.

Tumor-specific responses show more dynamic changes in nonresponders

Last, we wanted to understand whether there were any unique dynamic changes of these predictive TCRs. To do this, we first examined, at the clonotype level, the change in frequencies of TCR sequences present in the pre-treatment and on-treatment samples as a function of likelihood of being tumor versus virus specific (Fig. 5F). We noted that while the predicted virus-specific TCRs showed little change between nonresponders and responders, the tumor-specific TCRs found in the pre-treatment samples showed a marked decrease in frequency in nonresponders versus responders and the tumor-specific TCRs found in the post-treatment samples showed a marked increase in frequency in nonresponders versus responders. This finding suggests a more rapid turnover of tumor-versus viral-specific clones in nonresponders versus responders. When aggregating the change in frequencies over each patient, we further observed the same finding in that there was a higher turnover of tumor-specific clones in nonresponders versus responders (Fig. 5G), suggesting a futility of the tumor-specific response in nonresponders and a higher turnover of tumor-specific TCRs in these patients.

DISCUSSION

In this work, we sought to understand the T cell repertoire determinants of response to immunotherapy in the clinical setting along

with their potential antigen specificities. While there has been prior work in the field understanding the quantitative aspects of TCR repertoire (i.e., diversity, abundance, etc.), there has yet to be a study of the sequence motifs/concepts in the TCR repertoire that may predict response to immunotherapy. In this work, we use and expand on a previously described set of deep learning algorithms for TCR repertoire analysis to create models that not only predict clinical response but also allow us to understand and propose a biological model that explains the differences in TCR repertoire in responders/nonresponders.

Much of the prior work in the field of cancer immunology that attempts to understand the antigenic determinants of response to therapy often tackles the problem from the side of the presented epitope/antigen. Computational pipelines are set up to take whole-exome sequencing (WES) data and predict the presented epitope (7–9, 11, 12, 38). However, because of the many sequential steps/algorithms that are required to go from mutation to immunologically relevant epitope (i.e., expression, proteosomal cleavage, major histocompatibility complex binding, and T cell recognition), these pipelines suffer in accuracy (39–43). The benefit of interrogating the TCR sequences/repertoire directly is that this is a direct measurement of the antigen-specific response in an immune response. The hurdle, however, has been to understand the antigenic information encoded in the TCR sequence, and other than direct empirical validation of TCR clones, there has existed no high-throughput,

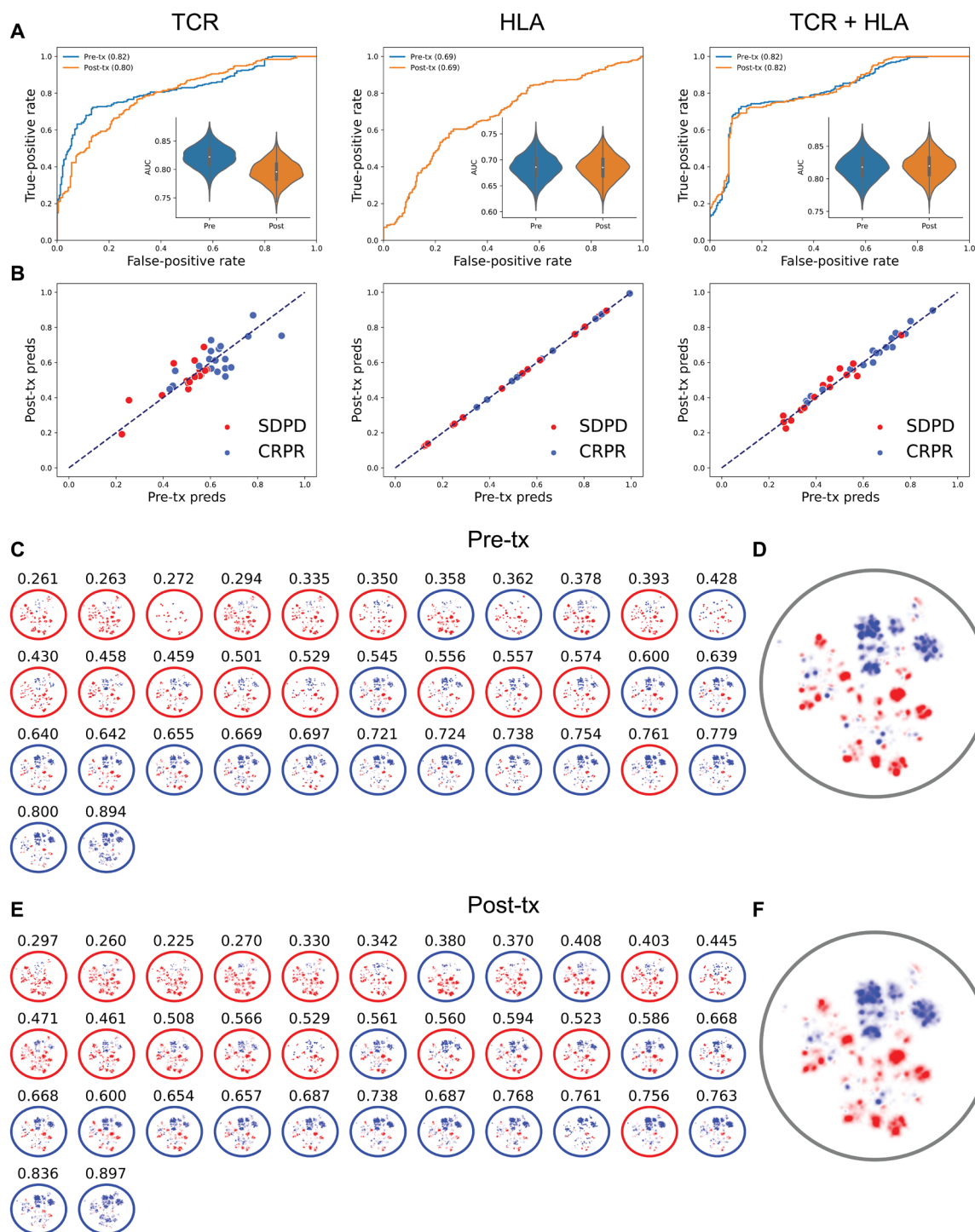


Fig. 4. Pre- versus post-treatment TCR repertoire. (A) All three respective models that were trained on pre-treatment repertoires were applied to 35 paired post-treatment TCR repertoires. Performance was then compared via visualizing ROC curves and bootstrapping the AUC to compare pre- versus post-treatment predictive signature. (B) For each pair of pre/post-treatment repertoires, the repertoire-level prediction was compared for pre- versus post-treatment across all trained models. To visualize distribution of predictive sequences, top 10% of predictive sequences in the 35 paired pre/postrepertoires were visualized across all paired samples (C and E) and over the entire cohort (D and F).

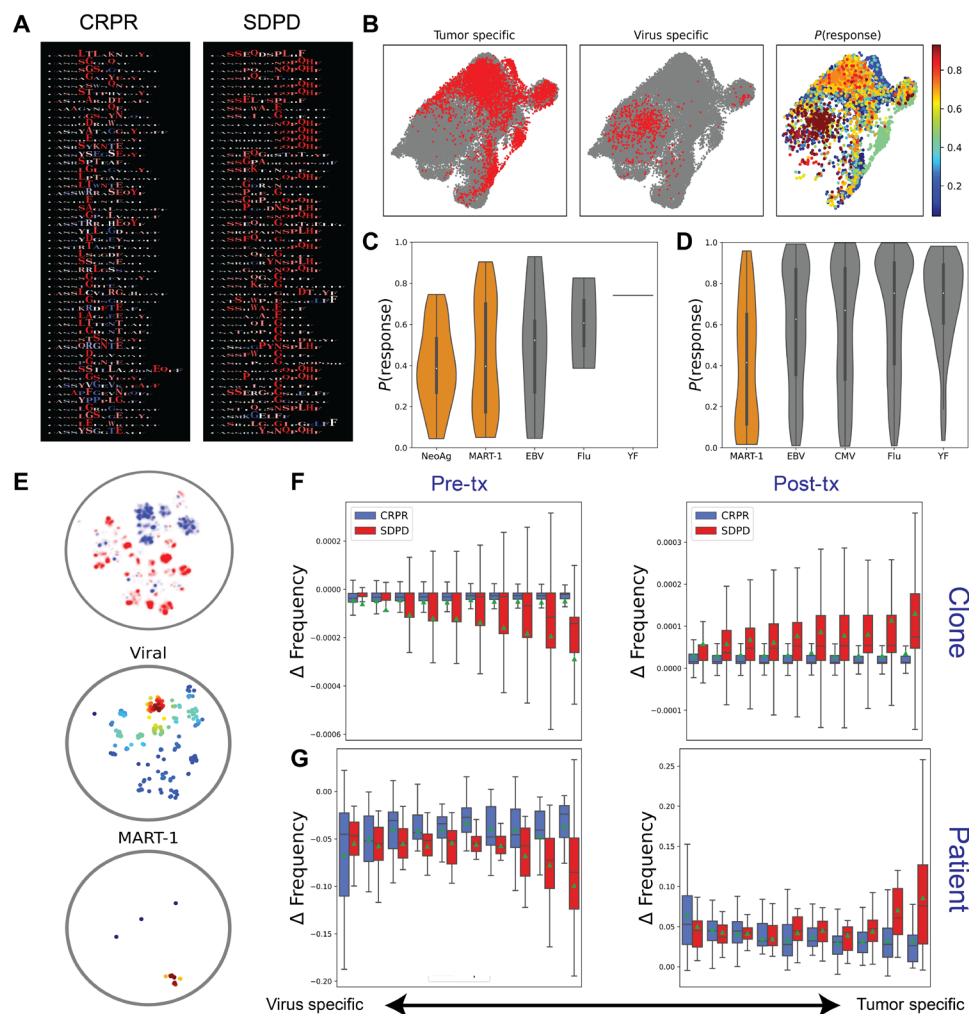


Fig. 5. Specificities and dynamics of predictive TCRs. (A) Residue sensitivity logos were created for the top 50 most predictive TCR sequences for responders (CRPR) and nonresponders (SDPD). (B) TCRs were collected from previous publication by Oliveria *et al.* (36) and are shown in phenotypic (derived from single-cell RNA sequencing) UMAP space and highlighted in red by antigen specificity (tumor- versus virus-specific TCRs), along with corresponding likelihood of response as determined by trained DeepTCR repertoire classifier. Likelihood of response distributions is shown for each class of antigen for the (C) Oliveria *et al.* (36) dataset and the (D) McPas-TCR database (orange, tumor specific; gray, virus specific). (E) Viral-specific versus MART-1-specific TCRs from McPas-TCR database present in the predictive TCRs from the CheckMate-038 cohort are shown within the unsupervised TCR sequence space. Color corresponds to kernel density estimate for that point in sequence space. (F) Clonotype-specific changes in frequency between pre- and post-treatment are shown for TCRs in pre-tx and post-tx samples as a function of likelihood of being virus- versus tumor-specific. Each set of paired box plots represents probability ranges of 10% (i.e., first pair of box plots represents sequences with probability of 0 to 10% of being tumor specific). Mean values for each box plot are denoted with a green triangle. (G) Changes in frequency of sequences in each probability bin were aggregated over each patient and shown as a function of likelihood of being virus versus tumor specific, showing the net change in frequency of a given set of sequences within each patient.

efficient method to interrogate the antigen specificity of the TCR sequence. Thus, there has been an effort in the machine learning domain to attempt to extract this antigenic information from the TCR sequence including methods such as DeepTCR (used in this work) (22, 23, 44–50). While the field is still nascent, with more and more data to train these models, they have the potential to completely change how we understand the antigen specificity of immune response directly from the TCR repertoire, thus avoiding highly variable and inaccurate prediction methods that attempt to predict the relevant epitopes. In this work, we demonstrate how methods such as DeepTCR would be used in the future to not only create possible predictive biomarkers in cancer but also to extract meaningful biological insights from TCR repertoire.

In the first part of this study, we first expand on our previous work by integrating HLA into the representation of the TCR sequence. While the TCR sequence can be thought of as containing the information required to understand antigen specificity, it really contains the information about the antigen/epitope in context of HLA. Given the high level of heterogeneity of the HLA alleles in the human population, a TCR sequence cannot be guaranteed to respond to the same epitope/antigen in different individuals. Therefore, we created a method to create a joint representation of TCR sequence and HLA background within which it was observed in. This joint representation then becomes a more complete and reliable measure of the epitope and allows direct comparison of TCR repertoire between HLA-mismatched individuals. When applying this approach to predicting response,

we do find that combining TCR sequence information and HLA background does improve predictive power of the model.

While the predictive power of the model is a key advantage of our approach in that we are able to aggregate the sum of TCR information in an entire repertoire to predict relevant information about response to therapy, much of the work presented focused on the interpretability of the model with hopes to reveal previously unappreciated biological insights. We first use a VAE, a completely unsupervised method of TCR sequence representation to characterize and visualize the predictive signature of response. When doing this, we find that our supervised MIL model did in fact pull out the relevant predictive signature from the background “noise” of TCR repertoire. In using an orthogonal method of validation with an unsupervised approach, this gave us further evidence that our supervised model had not become overfit to the data, and when looking at the distribution of the predictive sequences among each patient, we were able to observe the nature of a conserved TCR sequence signature in responders/nonresponders that was shared among multiple patients. This led us to ask the inevitable question about what could be the specificities of these predictive TCR sequences. By using two previously published datasets with known TCR to specificity relationships, we found that responders were enriched with a predictive signature that resembled a viral response, while nonresponders were enriched with a signature that resembled a tumor-specific response. While initially unexpected, we reasoned that the viral signature represents a background T cell response within the tumor [as has been validated in other studies (36)], and an accumulation of tumor-specific T cells in the nonresponders is relative to this background viral signature. Along with the previously published dataset that linked TCR sequence not only to antigen specificity but also to phenotype, we reasoned that this accumulation of tumor-specific T cells represents terminally differentiated effector T cells that may have become dysfunctional, and hence, their accumulation in the nonresponders.

When studying the dynamics of these antigen-specific responses before and after initiation of immunotherapy, while the antigen-specific signature did not change on therapy, we were surprised to see that there was a higher turnover of tumor-specific T cells in the nonresponders. Taken all these observations together, we propose a biological model of the dynamics and antigen-specific characteristics under immunotherapy and how these characteristics differ between responders and nonresponders to therapy (Fig. 6). Notably, nonresponders are characterized by an accumulation of dysfunctional tumor-specific T cells that undergo a higher level of turnover when under immunotherapy treatment, suggesting a futility of the ongoing T cell response to the tumor. In contrast, responders maintain the existing tumor-specific response within the tumor and their function is rescued by immunotherapy treatment and thus, the T cells already present in the tumor are able to be effective in their antitumor activity.

Last, this biological model is consistent with findings already reported in previous transcriptomic studies in the field. In the study by Oliveira *et al.* (36), nonresponder patients were characterized by an accumulation of high levels of putative tumor-reactive T cells, which were significantly higher in nonresponder patients with melanoma. Such specificities, even if at high frequencies, were not able to confer a productive antitumor response, because of the high level of exhaustion, as measured within the tumor microenvironment through single-cell RNA (36). In line with this view, patients with melanoma with response to checkpoint blockade are characterized

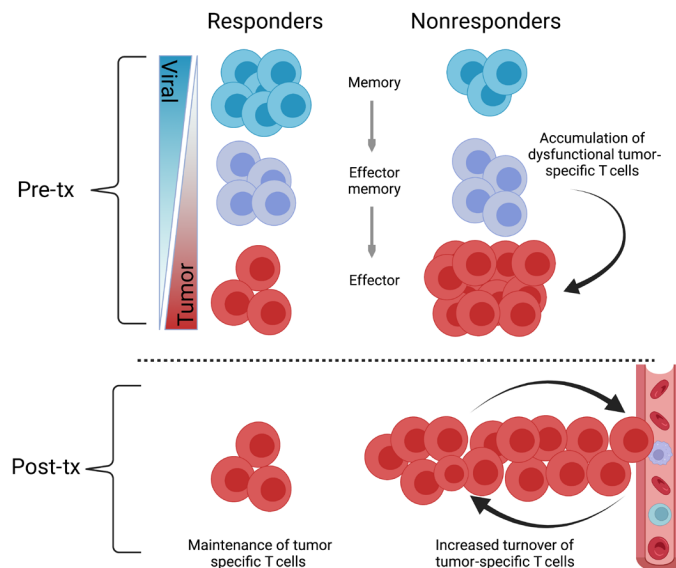


Fig. 6. Dynamics of tumor-specific T cells under immunotherapy. In the pre-treatment setting, nonresponders to immunotherapy have accumulated tumor-specific dysfunctional effector T cells. Following initiation of immunotherapy, nonresponders undergo a higher rate of turnover of tumor-specific T cells as compared with responders. Created with BioRender.com.

higher proportion of putative virus-specific T cells within the tumor specimens, while nonresponders are characterized by the accumulation of exhausted tumor-infiltrating lymphocytes (35, 36).

While the findings of this study demonstrate both a method by which to apply interpretable machine learning to TCR repertoire analysis and the biological insights one could appreciate, there are certainly limitations to this work. The most significant limitation of the study is the small size of the training/validation cohorts used in this study. Deep learning models are notorious for their ability to overfit to data, and there is much consideration in training these models so they do not overfit to spurious or irrelevant information. To address this major limitation, we made sure to only assess performance of the model in the test set during cross-validation. Furthermore, by corroborating the finding of this predictive sequence signature with a VAE, a completely unsupervised method, we were able to provide further evidence that our supervised model had not overfit to the data. Last, we validated the predictive signature in the original CheckMate-038 cohort in two additional clinical cohorts undergoing treatment with checkpoint blockade, providing further validation of the observed finding.

When taken together, these findings highlight the utility of methods in deep learning to identify key features of specificity in TCR repertoire and their dynamics under the influence of immunotherapy and how they relate to clinical response. Further work in this area might be able to leverage these described approaches for biomarker development as well as aid in the understanding and development of better targeted treatments in the age of precision oncology.

METHODS

CheckMate-038 experimental model and participant details

CheckMate-038 is a multi-arm, multi-institutional, institutional review board–approved, prospective study (CA209-038; NCT01621490).

Patients in parts 2 to 4 received either nivolumab (3 mg/kg) every 2 weeks ($n = 21$) or nivolumab (1 mg/kg) + ipilimumab (3 mg/kg) every 3 weeks $\times$ 4 doses and then nivolumab (3 mg/kg) every 2 weeks ($n = 62$), until progression or for a maximum of 2 years. Radiographic assessment of response was performed approximately every 8 weeks until progression. Progression was confirmed with a repeat computed tomography scan typically 4 weeks later. Tumor response for patients was defined by RECIST v1.1. Response to therapy indicates best overall response unless otherwise indicated. All patients underwent biopsy of a metastatic lesion before commencing therapy (1 to 7 days before first dose of therapy). Tumor tissue was divided for formalin-fixation, paraffin embedding (FFPE) or storage with RNA later (Ambion) for subsequent RNA/DNA extraction. PD-L1 expression on the tumor cell surface was assessed in FFPE samples at a central laboratory (Dako 28-8 antibody). The clinical trial protocol and its amendments were approved by the relevant institutional review boards, and the study was conducted in accordance with the Declaration of Helsinki and the International Conference on Harmonization Guidelines for Good Clinical Practice. All patients signed written informed consent before having any study procedures performed.

CheckMate-038 TCR-seq and HLA data generation

Tumor biopsy samples were collected before initiation of therapy and stored in RNA later. DNA was extracted and submitted to Adaptive Biotechnologies for survey-level TCR β -chain sequencing, where targeted amplicon libraries were prepared by multiplex polymerase chain reaction targeting all TCR β -chain V/D/J gene segments and sequenced using the Illumina HiSeq system (51, 52). Data for individual TCR sequences, previously analyzed by Anagnostou *et al.* (21), including V/D/J gene segment identification and CDR3- β sequence, were obtained from Adaptive Biotechnologies for analysis via DeepTCR. Tumor biopsy DNA was also sent for WES (Personal Genome Diagnostics) to determine TMB, and OptiType was used to infer HLA genotype of the patients (53). Data from patients who consented to deposition will be submitted to the European Genome-phenome Archive (21).

Data curation

TCR-seq files were collected as raw tsv/csv-formatted files from the various sources cited within the manuscript. Sequencing files were parsed to take the amino acid sequence of the CDR3 after removing unproductive sequences. Clones with different nucleotide sequences but the same amino acid sequence were aggregated together under one amino acid sequence, and their reads were summed to determine their relative abundance. Within the parsing code, we additionally specified to ignore sequences that used non-International Union of Pure and Applied Chemistry letters (*, X, and O) and removed sequences that were greater than 40 amino acids in length. For the purpose of the algorithm, the maximum length can be altered, but we chose 40 as we did not expect any real sequences to be longer than this length.

Training DeepTCR repertoire classifier

To identify a predictive signature of response in the TCR repertoire of the tumor microenvironment before initiation of therapy, we used DeepTCR (v2.1.6), a deep learning framework for revealing sequence concepts within T cell repertoires (22). We make one significant change to the existing software by allowing the incorporation

of HLA information within the representation of the TCR. This is accomplished by representing the HLA background that a given TCR is observed within as a categorical multihot encoded variable as an input for the neural network. All other aspects of the approach are as initially described in the original manuscript in which DeepTCR was first presented. Notably, we fit the repertoire classifier on the CheckMate-038 data using TCR sequence information (CDR3- β and V/D/J), HLA, or TCR + HLA information provided to demonstrate the different types of information each of the inputs contributes to the predictive power of the model. For each type of input that was tested, the same exact train/test splits during MC cross-validation were used as to allow fair comparisons when comparing models trained with different input data. Furthermore, because of the small nature of the CheckMate-038 dataset, training had to be accomplished in a way to prevent overfitting of the repertoire classifier. Therefore, to train the repertoire classifier on these datasets, we used an MC cross-validation where in the hinge loss was used during model training, which prevents the model from further reducing the loss of any given sample below a defined threshold. The idea behind this type of objective function is that once a sample has been predicted sufficiently correct, the network is not encouraged to drop its loss any further and thus reduces overfitting on the training data. Model training with this hinge loss was stopped once a predefined threshold was met, and in keeping MC cross-validation, model performance was assessed on the testing data of that train/test split. We then used a bootstrapping method where we sampled the MC predictions with replacement 5000 times to approximate a confidence interval around the AUC. All hyperparameters for the DeepTCR models can be found in the publicly available GitHub repository as provided below in Acknowledgments (Data and materials availability).

Validation cohorts

TCR-seq data were collected from two previously published manuscripts denoted as yost (34) and sade (35), which consisted of patients with basal/squamous cell carcinoma and melanoma, respectively. The yost dataset consisted of samples from 11 patients who had TCR-seq available from pre-treatment biopsies and provided available on immuneACCESS, and the sade dataset consisted of samples from 19 patients who had TCR-seq available from pre-treatment biopsies and provided in the original publication materials. Both of these cohorts consisted of patients who were treated with checkpoint blockade therapy and were assessed for clinical response to treatment via RECIST criteria, in a similar fashion as was performed within the CheckMate-038 cohort. The DeepTCR repertoire classifiers that fit to the CheckMate-038 cohort were then used to conduct repertoire level inference on the patients within these two independent clinical cohorts, and predictive performance was assessed via ROC and AUC measurements.

Unsupervised representations via VAE and UMAP

To provide interpretability of the predictive signature found, we used the DeepTCR VAE to do unsupervised dimensionality reduction of all TCRs found in the CheckMate-038 cohort. Each instance into the VAE is defined by a joint representation of CDR3- β , V/D/J gene usage, and the HLA background the TCR was found within as previously described. Using the VAE, this input is transformed into a 128-dimensional latent vector before being further reduced to two dimensions via UMAP (default settings for python package

umap-learn). For the purpose of visualization, because each TCR also has a corresponding frequency associated with it, this information was used to construct two-dimensional histograms to visualize the density of these TCRs in the UMAP latent space.

Post-treatment inference

To apply models from the pre-treatment cohort to the post-treatment cohort, we used a method to prevent overinflation of performance characteristics because the pre- and post-treatment samples are highly related (coming from the same patients). To do this, we only used models on individuals in the post-treatment setting that were not trained on those individuals' pre-treatment tumors. In other words, when the model was trained on a given partition of the pre-treatment data and then was tested on the other partition (test set) of the pre-treatment data and the paired-test set's post-treatment data. This type of cross-validation prevented the model from making a prediction on a patient it had been trained on, whether it be trained on pre- or post-treatment samples.

Linking predictive model to known antigen specificity

To interrogate the antigen specificity of the predictive signatures of response/nonresponse, we collected two previously published datasets that have empiric validation of CDR3 sequences to antigens/epitopes. As our clinical cohort consists of patients with melanoma, we first use a melanoma relevant dataset where the authors established links between TCR sequence, antigen specificity, and gene expression phenotype (36). We also use McPas-TCR, a larger dataset with TCR sequences and their known specificities (37). For the melanoma dataset (36), because this dataset had the CDR3- β sequence, V/D/J gene usage, and HLA background of the individuals, we were able to score each TCR by running it through our pretrained model. In our analysis of the McPas-TCR database, to maximize the overlap of TCRs between our cohort of patients and those found in the database, we cross-matched TCRs from our clinical cohort (with predictive likelihoods) with TCRs in the McPas-TCR database at only the CDR3- β sequence level to assign known antigen-specific TCRs to their likelihood of response/nonresponse.

Clonal dynamics as a function of likelihood of response

In the CheckMate-038 cohort, because there exists paired pre- and post-treatment biopsies for a subset of the patients, we wanted to interrogate the clonal dynamics in light of information provided by the predictive model of response. To do this, we divided all the TCR sequences into 10 ordinal categories representing the spectrum of virus- to tumor-specific TCRs as predicted by our model. We then further stratified these sequences by whether they were in responders (CRPR) or nonresponders (SDPD). We then looked at their clonal dynamics in reference to either their pre- or post-treatment frequencies. For the TCR sequences seen in the pre-treatment biopsies, we looked at their change in frequencies post-treatment with respect to their pre-treatment frequencies, and for the TCR sequences seen in the post-treatment biopsies, we looked at their change in frequencies pre-treatment with respect to their post-treatment frequencies. To further quantify the dynamics of TCRs at the sample/patient level, we aggregated the change in frequencies over each patient across the respective bins along the spectrum of virus- to tumor-specific to output a net change in frequency per patient.

Statistical tests and machine learning models

All statistical tests applied to data were implemented with the scipy.stats module. Classical machine learning techniques and performance metrics were implemented with scikit-learn.

SUPPLEMENTARY MATERIALS

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